

NE504 - Nuclear fuel cycle analysis

Xenon neutron poison production

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1 Mathematical models

1.1 Iodine

Rate of change of iodine -

$$\begin{aligned}\frac{dI}{dt} &= \gamma_I \Sigma_F \phi_T - \lambda_I I \\ I(0) &= 0\end{aligned}\tag{1}$$

1.2 Xenon

Rate of change of xenon -

$$\begin{aligned}\frac{dX}{dt} &= \lambda_I I + \gamma_X \Sigma_F \phi_T - \lambda_X X - \sigma_A \phi_T X \\ X(0) &= 0\end{aligned}\tag{2}$$

2 Solutions

2.1 Iodine

Apply laplace transform to Eq. (1) -

$$\begin{aligned}\frac{dI}{dt} &= \gamma_I \Sigma_F \phi_T - \lambda_I I \\ s\tilde{I} &= \frac{1}{s} \gamma_I \Sigma_F \phi_T - \lambda_I \tilde{I}\end{aligned}\tag{3}$$

Rearrange terms to obtain laplace solution -

$$\tilde{I} = \frac{1}{s(s + \lambda_I)} \gamma_I \Sigma_F \phi_T\tag{4}$$

Invert Eq. (4) to obtain real time domain solution -

$$I(t) = \frac{\gamma_I \Sigma_F \phi_T}{\lambda_I} (1 - e^{-\lambda_I t})\tag{5}$$

2.2 Xenon

Substitute the iodine solution in Eq. (5) into Eq. (2) -

$$\begin{aligned}\frac{dX}{dt} &= \lambda_I I + \gamma_X \Sigma_F \phi_T - \lambda_X X - \sigma_A \phi_T X \\ \frac{dX}{dt} &= \lambda_I \frac{\gamma_I \Sigma_F \phi_T}{\lambda_I} (1 - e^{-\lambda_I t}) + \gamma_X \Sigma_F \phi_T - \lambda_X X - \sigma_A \phi_T X\end{aligned}\tag{6}$$

Rearrange the terms and simplify -

$$\begin{aligned}\frac{dX}{dt} &= \gamma_I \Sigma_F \phi_T (1 - e^{-\lambda_I t}) + \gamma_X \Sigma_F \phi_T - \lambda_X X - \sigma_A \phi_T X \\ \frac{dX}{dt} &= \gamma_I \Sigma_F \phi_T - \gamma_I \Sigma_F \phi_T e^{-\lambda_I t} + \gamma_X \Sigma_F \phi_T - \lambda_X X - \sigma_A \phi_T X\end{aligned}\tag{7}$$

$$\frac{dX}{dt} = \gamma_I \Sigma_F \phi_T - \gamma_I \Sigma_F \phi_T e^{-\lambda_I t} + \gamma_X \Sigma_F \phi_T - (\lambda_X + \sigma_A \phi_T) X\tag{8}$$

Apply laplace transform to Eq. (8) -

$$s\tilde{X} = \frac{1}{s} \gamma_I \Sigma_F \phi_T - \frac{1}{s + \lambda_I} \gamma_I \Sigma_F \phi_T + \frac{1}{s} \gamma_X \Sigma_F \phi_T - (\lambda_X + \sigma_A \phi_T) \tilde{X}\tag{9}$$

Rearrange the terms in Eq. (9) to obtain the laplace solution -

$$(s + [\lambda_X + \sigma_A \phi_T]) \tilde{X} = \frac{1}{s} \gamma_I \Sigma_F \phi_T - \frac{1}{s + \lambda_I} \gamma_I \Sigma_F \phi_T + \frac{1}{s} \gamma_X \Sigma_F \phi_T\tag{10}$$

Then -

$$\tilde{X} = + \frac{1}{s(s + [\lambda_X + \sigma_A \phi_T])} \gamma_I \Sigma_F \phi_T - \frac{1}{(s + \lambda_I)(s + [\lambda_X + \sigma_A \phi_T])} \gamma_I \Sigma_F \phi_T + \frac{1}{s(s + [\lambda_X + \sigma_A \phi_T])} \gamma_X \Sigma_F \phi_T\tag{11}$$

Invert Eq. (11) to obtain real time domain solution -

$$X(t) = +\frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} (1 - e^{-(\lambda_X + \sigma_A \phi_T)t}) - \frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} (e^{-\lambda_I t} - e^{-(\lambda_X + \sigma_A \phi_T)t}) + \frac{\gamma_X \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} (1 - e^{-(\lambda_X + \sigma_A \phi_T)t}) \quad (12)$$

3 Equilibrium time

The equilibrium time for xenon should be when the concentration achieves a steady state; i.e., the rate of change in the concentration is zero.

Then, compute the derivative of the xenon concentration model in Eq. (12) -

$$\begin{aligned} \frac{dX}{dt} = & + \left[\frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} \right] ((\lambda_X + \sigma_A \phi_T) e^{-(\lambda_X + \sigma_A \phi_T)t}) \\ & - \left[\frac{\gamma_I \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T - \lambda_I} \right] ((\lambda_X + \sigma_A \phi_T) e^{-(\lambda_X + \sigma_A \phi_T)t} - \lambda_I e^{-\lambda_I t}) \\ & + \left[\frac{\gamma_X \Sigma_F \phi_T}{\lambda_X + \sigma_A \phi_T} \right] ((\lambda_X + \sigma_A \phi_T) e^{-(\lambda_X + \sigma_A \phi_T)t}) \end{aligned} \quad (13)$$

Multiply out all the terms -

$$\begin{aligned} \frac{dX}{dt} = & + \gamma_I \Sigma_F \phi_T e^{-(\lambda_X + \sigma_A \phi_T)t} \\ & - \frac{(\gamma_I \Sigma_F \phi_T)(\lambda_X + \sigma_A \phi_T)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)} e^{-(\lambda_X + \sigma_A \phi_T)t} \\ & + \frac{(\gamma_I \Sigma_F \phi_T)(\lambda_I)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)} e^{-\lambda_I t} \\ & + \gamma_X \Sigma_F \phi_T e^{-(\lambda_X + \sigma_A \phi_T)t} \end{aligned} \quad (14)$$

Let $\frac{dX}{dt} = 0$ and group the terms by exponential -

$$(\gamma_I \Sigma_F \phi_T + \gamma_X \Sigma_F \phi_T - \frac{(\gamma_I \Sigma_F \phi_T)(\lambda_X + \sigma_A \phi_T)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)})e^{-(\lambda_X + \sigma_A \phi_T)t} + \frac{(\gamma_I \Sigma_F \phi_T)(\lambda_I)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)}e^{-\lambda_I t} = 0 \quad (15)$$

The next series of steps is just some algebra -

$$\begin{aligned} -(\gamma_I \Sigma_F \phi_T + \gamma_X \Sigma_F \phi_T - \frac{(\gamma_I \Sigma_F \phi_T)(\lambda_X + \sigma_A \phi_T)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)})e^{-(\lambda_X + \sigma_A \phi_T)t} &= \frac{(\gamma_I \Sigma_F \phi_T)(\lambda_I)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)}e^{-\lambda_I t} \\ (-\gamma_I \Sigma_F \phi_T - \gamma_X \Sigma_F \phi_T + \frac{(\gamma_I \Sigma_F \phi_T)(\lambda_X + \sigma_A \phi_T)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)})e^{-(\lambda_X + \sigma_A \phi_T)t} &= \frac{(\gamma_I \Sigma_F \phi_T)(\lambda_I)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)}e^{-\lambda_I t} \end{aligned} \quad (16)$$

$$(\frac{(\gamma_I \Sigma_F \phi_T)(\lambda_X + \sigma_A \phi_T)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)} - \gamma_I \Sigma_F \phi_T - \gamma_X \Sigma_F \phi_T)e^{-(\lambda_X + \sigma_A \phi_T)t} = \frac{(\gamma_I \Sigma_F \phi_T)(\lambda_I)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)}e^{-\lambda_I t} \quad (17)$$

Divide through Eq. (17) by $e^{-\lambda_I t}$ and $(\frac{(\gamma_I \Sigma_F \phi_T)(\lambda_X + \sigma_A \phi_T)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)} - \gamma_I \Sigma_F \phi_T - \gamma_X \Sigma_F \phi_T)$ -

$$\frac{e^{-(\lambda_X + \sigma_A \phi_T)t}}{e^{-\lambda_I t}} = \frac{\frac{(\gamma_I \Sigma_F \phi_T)(\lambda_I)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)}}{(\frac{(\gamma_I \Sigma_F \phi_T)(\lambda_X + \sigma_A \phi_T)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)} - \gamma_I \Sigma_F \phi_T - \gamma_X \Sigma_F \phi_T)} \quad (18)$$

Compute the exponential division -

$$e^{(\lambda_I - (\lambda_X + \sigma_A \phi_T))t} = \frac{\frac{(\gamma_I \Sigma_F \phi_T)(\lambda_I)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)}}{(\frac{(\gamma_I \Sigma_F \phi_T)(\lambda_X + \sigma_A \phi_T)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)} - \gamma_I \Sigma_F \phi_T - \gamma_X \Sigma_F \phi_T)} \quad (19)$$

Compute the natural logarithm -

$$(\lambda_I - (\lambda_X + \sigma_A \phi_T))t = \ln\left[\frac{\frac{(\gamma_I \Sigma_F \phi_T)(\lambda_I)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)}}{(\frac{(\gamma_I \Sigma_F \phi_T)(\lambda_X + \sigma_A \phi_T)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)} - \gamma_I \Sigma_F \phi_T - \gamma_X \Sigma_F \phi_T)}\right] \quad (20)$$

Solve for time (t) -

$$t = \frac{1}{\lambda_I - (\lambda_X + \sigma_A \phi_T)} \cdot \ln \left[\frac{\frac{(\gamma_I \Sigma_F \phi_T)(\lambda_I)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)}}{\left(\frac{(\gamma_I \Sigma_F \phi_T)(\lambda_X + \sigma_A \phi_T)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)} - \gamma_I \Sigma_F \phi_T - \gamma_X \Sigma_F \phi_T \right)} \right] \quad (21)$$

Simplify $\ln[\cdot]$ -

$$\frac{\frac{(\gamma_I \Sigma_F \phi_T)(\lambda_I)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)} \cdot (\lambda_X + \sigma_A \phi_T - \lambda_I)}{\left(\frac{(\gamma_I \Sigma_F \phi_T)(\lambda_X + \sigma_A \phi_T)}{(\lambda_X + \sigma_A \phi_T - \lambda_I)} \cdot (\lambda_X + \sigma_A \phi_T - \lambda_I) - \gamma_I \Sigma_F \phi_T (\lambda_X + \sigma_A \phi_T - \lambda_I) - \gamma_X \Sigma_F \phi_T (\lambda_X + \sigma_A \phi_T - \lambda_I) \right)} \quad (22)$$

$$\frac{\gamma_I \lambda_I \Sigma_F \phi_T}{(\gamma_I \Sigma_F \phi_T)(\lambda_X + \sigma_A \phi_T) - \gamma_I \Sigma_F \phi_T (\lambda_X + \sigma_A \phi_T - \lambda_I) - \gamma_X \Sigma_F \phi_T (\lambda_X + \sigma_A \phi_T - \lambda_I)} \quad (23)$$

$$\frac{\gamma_I \lambda_I \Sigma_F \phi_T}{(\gamma_I \Sigma_F \phi_T)(\lambda_X + \sigma_A \phi_T) - (\gamma_I \Sigma_F \phi_T)(\lambda_X + \sigma_A \phi_T - \lambda_I) - (\gamma_X \Sigma_F \phi_T)(\lambda_X + \sigma_A \phi_T - \lambda_I)} \quad (24)$$

$$\frac{\gamma_I \lambda_I \Sigma_F \phi_T}{(\gamma_I \Sigma_F \phi_T)((\lambda_X + \sigma_A \phi_T) - (\lambda_X + \sigma_A \phi_T - \lambda_I)) - (\gamma_X \Sigma_F \phi_T)(\lambda_X + \sigma_A \phi_T - \lambda_I)} \quad (25)$$

$$\frac{\gamma_I \lambda_I \Sigma_F \phi_T}{(\gamma_I \Sigma_F \phi_T)(\gamma_X + \sigma_A \phi_T - \gamma_X - \sigma_A \phi_T + \lambda_I) - (\gamma_X \Sigma_F \phi_T)(\lambda_X + \sigma_A \phi_T - \lambda_I)} \quad (26)$$

$$\frac{\gamma_I \lambda_I \Sigma_F \phi_T}{\gamma_I \lambda_I \Sigma_F \phi_T - (\gamma_X \Sigma_F \phi_T)(\lambda_X + \sigma_A \phi_T - \lambda_I)} \quad (27)$$

$$\frac{\gamma_I \lambda_I \Sigma_F(\phi_T)}{\gamma_I \lambda_I \Sigma_F(\phi_T) - (\gamma_X \Sigma_F(\phi_T))(\lambda_X + \sigma_A \phi_T - \lambda_I)} \quad (28)$$

$$\frac{\gamma_I \lambda_I \Sigma_F}{\gamma_I \lambda_I \Sigma_F - \gamma_X \Sigma_F(\lambda_X + \sigma_A \phi_T - \lambda_I)} \quad (29)$$

Finally, substitute Eq. (29) back into Eq. (30) -

$$t = \frac{1}{\lambda_I - (\lambda_X + \sigma_A \phi_T)} \cdot \ln\left[\frac{\gamma_I \lambda_I \Sigma_F}{\gamma_I \lambda_I \Sigma_F - \gamma_X \Sigma_F(\lambda_X + \sigma_A \phi_T - \lambda_I)}\right] \quad (30)$$